AREA AND PERIMETER, (AREA OF RECTANGLES, TRIANGLES AND

Name: ____________________ **Date:** ______________

LO: I can calculate the area of rectangles, triangles and parallelograms.

Area formulae

Area is the amount of space inside a 2D shape, measured in **square units** (cm^2, m^2).

Formulae: Rectangle: A = l × w. Triangle: A = ½ × b × h. Parallelogram: A = b × h (perpendicular height).

Key facts

●	Rectangle: A = length × width	l × w
●	Triangle: A = ½ × base × height	half of rectangle
●	Parallelogram: A = base × height	perpendicular h
●	Triangle: A = base × height	forgot ½
●	Parallelogram: use slant	must be

Key idea

Always use the PERPENDICULAR height (not the slant). Triangle is half of the enclosing rectangle.

A = ½ × b × h

Triangle: half base times height

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – rectangle

Find the area of a rectangle 8 cm by 5 cm.

A = l × w

A = 8 × 5

A = 40 cm^2

= 40 cm^2

Multiply length by width.

Example 2 – triangle

Find the area of a triangle with base 10 cm and height 6 cm.

A = ½ × b × h

A = ½ × 10 × 6

A = 30 cm^2

= 30 cm^2

Half of base times height.

Example 3 – parallelogram

A parallelogram has base 12 cm and perpendicular height 7 cm. Find its area.

A = b × h

A = 12 × 7

A = 84 cm^2

= 84 cm^2

Use the perpendicular height, not the slant side.

Example 4 – working backwards

A triangle has area 24 cm^2 and base 8 cm. Find the height.

A = ½ × b × h

24 = ½ × 8 × h   24 = 4h

h = 6 cm

h = 6 cm

Rearrange the formula: h = 2A ÷ b.

YOUR TURN – PRACTICE (deliberate practice)

1. Area of rectangles
Find the area.

a) l = 7 cm, w = 4 cm

b) l = 12 cm, w = 5 cm

c) l = 9 cm, w = 9 cm

d) l = 15 cm, w = 3 cm

2. Area of triangles
Find the area. Use $A = \frac{1}{2} \times b \times h$.

a) b = 6 cm, h = 4 cm

b) b = 10 cm, h = 8 cm

c) b = 14 cm, h = 5 cm

d) b = 9 cm, h = 12 cm

3. Area of parallelograms
Find the area. Use perpendicular height.

a) b = 8 cm, h = 6 cm

b) b = 11 cm, h = 4 cm

c) b = 15 cm, h = 9 cm

d) b = 7.5 cm, h = 8 cm

4. Finding missing measurements
Find the missing length.

a) Rectangle: A = 36 cm^2, l = 9 cm. Find w.

b) Triangle: A = 20 cm^2, b = 8 cm. Find h.

c) Parallelogram: A = 72 cm^2, h = 8 cm. Find b.

d) Triangle: A = 30 cm^2, h = 10 cm. Find b.

5. Mixed area problems
Calculate each area.

a) Rectangle: 6.5 cm by 4 cm

b) Triangle: b = 16 cm, h = 7 cm

c) Parallelogram: b = 20 cm, h = 3.5 cm

d) Square: side 11 cm

e) Triangle: b = 5 cm, h = 5 cm

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Area of a triangle = $\frac{1}{2}$ × base × height ☐

Correct answer: __________

Reason: _________________

b) Area of a triangle = base × height ☐

Correct answer: __________

Reason: _________________

c) Parallelogram area uses perpendicular height ☐

Correct answer: __________

Reason: _________________

d) Area is measured in cm ☐

Correct answer: __________

Reason: _________________

e) A square is a special rectangle ☐

Correct answer: __________

Reason: _________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A rectangular garden is **12 m by 8 m**. A triangular flower bed has base **6 m** and height **4 m**. What fraction of the garden is the flower bed?

Expression: ____________________________

Simplified: ____________________________

b) A parallelogram and a triangle share the same **base of 10 cm** and the same **height of 6 cm**. What is the relationship between their areas? Will this always be true?

Expression: ____________________________

Simplified: ____________________________

TOP TIPS
Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK
Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!